

Calculate exactly with fractions and multiples of π where appropriate

Step 53

★ Today I am learning to calculate exactly with fractions and multiples of π .

1. Exact Fraction Calculations

Calculate each expression. Give your answer in simplest form.

a) $\frac{1}{2} + \frac{1}{3} =$ _____

b) $\frac{3}{4} - \frac{1}{6} =$ _____

c) $\frac{2}{5} \times \frac{15}{8} =$ _____

d) $\frac{7}{9} \div \frac{14}{27} =$ _____

e) $\frac{2}{3} + \frac{5}{6} - \frac{1}{4} =$ _____

f) $\frac{5}{6} \times \frac{3}{10} \div \frac{1}{4} =$ _____

g) $3 - \frac{5}{12} =$ _____

h) $2\frac{1}{4} - \frac{3}{8} =$ _____

Remember:

- Add or subtract: find a common denominator.
- Multiply: multiply across.
- Divide: multiply by the reciprocal.
- Always simplify your answer.

2. Fractions in Context

Solve each problem. Give your answer in simplest form.

 a) A recipe uses $\frac{2}{3}$ cup of sugar. You want to make $1\frac{1}{2}$ times the recipe. How much sugar do you need?

Answer: _____

 b) A ribbon is $\frac{3}{4}$ m long. It is cut into pieces of $\frac{1}{6}$ m. How many pieces can be made?

Answer: _____

 c) A tank is $\frac{2}{5}$ full of water. Then $\frac{1}{3}$ of the tank is added. What fraction of the tank is full now?

Answer: _____

 d) A bag has $\frac{5}{8}$ kg of rice. You use $\frac{3}{10}$ kg. How much rice is left in the bag?

Answer: _____

Think about:

- What operation(s) are needed?
- Write your work using fractions.
- Simplify your answer.

3. Exact Calculations with π

 Calculate each expression. Leave your answer in terms of π in simplest form.

a) $2\pi + 3\pi =$ _____

b) $7\pi - 4\pi =$ _____

c) $5\pi \times 3 =$ _____

d) $12\pi \div 4 =$ _____

e) $\frac{\pi}{2} + \frac{\pi}{3} =$ _____

f) $\frac{3\pi}{4} - \frac{\pi}{6} =$ _____

g) $\frac{2}{2}\pi \times 15 =$ _____

h) $\frac{2}{5}\pi \times 15 =$ _____

i) $\left(\frac{3}{2}\pi\right) \div \left(\frac{3}{4}\right) =$ _____

Remember:

- Combine like terms with π .
- Multiply or divide the numerical parts.
- Leave π in your answer.

4. Circumference and Area of Circles

 Give your answers in terms of π in simplest form.

 a) Find the circumference of a circle with radius 7 cm.
Answer: _____

 b) Find the area of a circle with radius 5 cm.
Answer: _____

 c) Find the circumference of a circle with diameter 18 cm.
Answer: _____

 d) Find the area of a circle with diameter 12 cm.
Answer: _____

Remember:

- Circumference: $C = 2\pi r$ or $C = \pi d$
- Area: $A = \pi r^2$
- Use the given information to choose the best formula.

5. Mixed Problems

Solve each problem. Give exact answers in simplest form.

 a) The perimeter of a square is 20π cm. What is the length of one side?

Answer: _____

 b) A sector of a circle with radius 10 cm has an arc length of 5π cm. What is the central angle of the sector in degrees?

Answer: _____

 c) A circular garden has area 36π m². Find the radius of the garden.

Answer: _____

 d) A wheel makes 15 complete revolutions. The wheel has a radius of 14 cm. How far does it travel? Give your answer in terms of π .

Answer: _____

Think about:

- What formula(s) do you need?
- Write your work clearly.
- Leave answers in simplest form.

6. Choose the Correct Expression

Write the correct expression and evaluate it. Give your answer in simplest form.

 a) Three-quarters of $\frac{2}{5}$

A $\frac{3}{4} \times \frac{2}{5}$

B $\frac{3}{4} + \frac{2}{5}$

C $\frac{2}{5} - \frac{3}{4}$

Answer: _____

 b) 5π less than $\frac{3}{2}\pi$

A $5\pi - \frac{3}{2}\pi$

B $\frac{3}{2}\pi - 5\pi$

C $5\pi + \frac{3}{2}\pi$

Answer: _____

 c) Twice the circumference of a circle with radius r

A $2 \times (2\pi r)$

B $2\pi \times (2r)$

C $2 \times (\pi r^2)$

Answer: _____

 d) The area of a circle with radius $3r$

A $\pi(3r)$

B $\pi(3r^2)$

C $\pi(3r)^2$

Answer: _____

Remember:

- Read carefully.
- Write the expression before you calculate.
- Keep answers exact.

7. Real-World Problems

Solve each problem. Give exact answers in simplest form.

 a) A pizza has a diameter of 30 cm. What is the area of the pizza?

Answer: _____

 b) A cylindrical can has a radius of 5 cm and height 12 cm. What is the total distance around the can (top and bottom together)?

Answer: _____

 c) A bike wheel has radius 35 cm. How many complete revolutions will the wheel make in 1 km? (Use $\pi = \frac{22}{7}$)

Answer: _____

 d) A circular swimming pool has circumference 44π m. How many meters of fencing are needed to go twice around the pool?

Answer: _____

Think about:

- Identify what is known.
- Choose the correct formula or operation.
- Show your work.

My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more practice.



I need help.

Parent Observation (optional)

Please tick the boxes that apply.

☐ Understands concepts

☐ Uses exact calculations

☐ Solves problems confidently

Notes: _____

☐ Explains thinking clearly

☐ Needs more practice

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____